CN & NP

SKILL 1

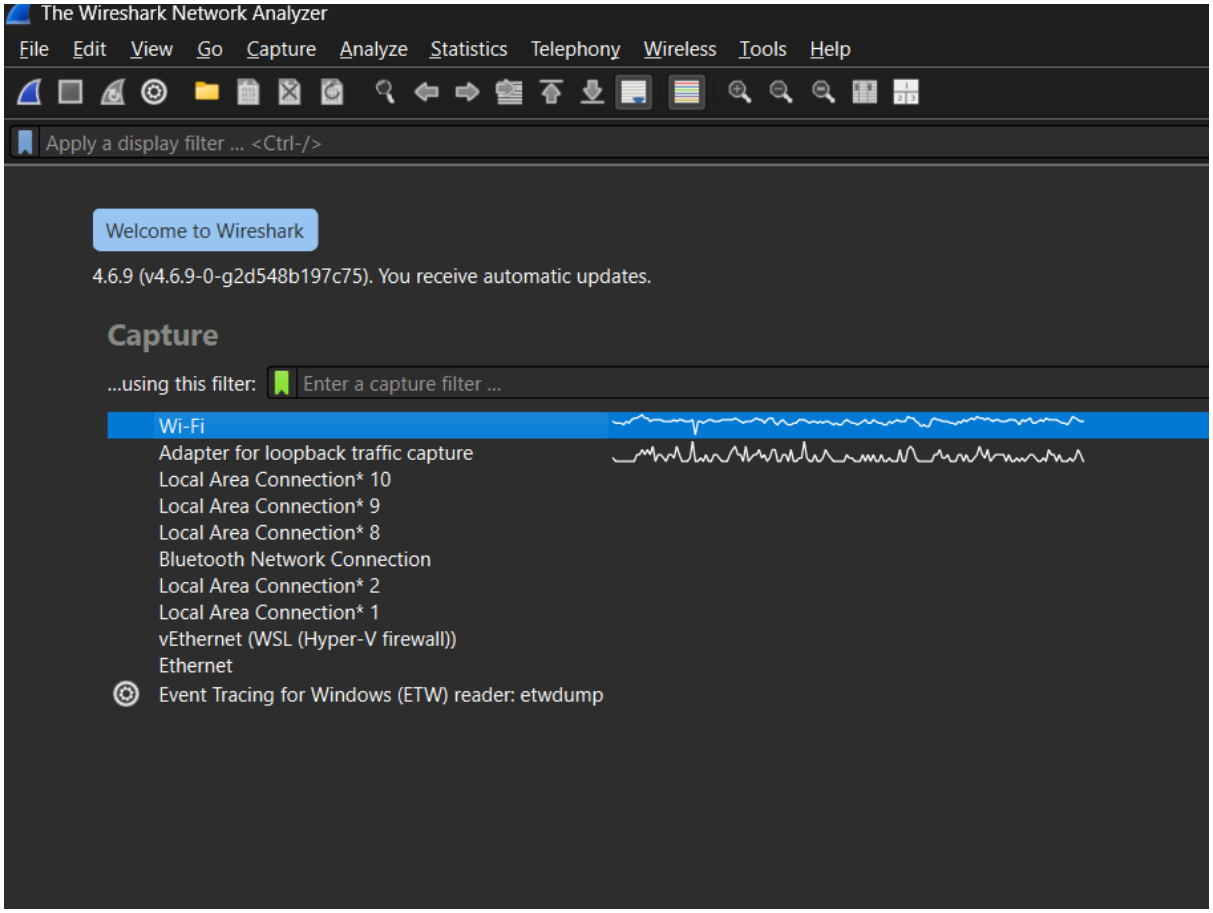
Roll no : 2520030101

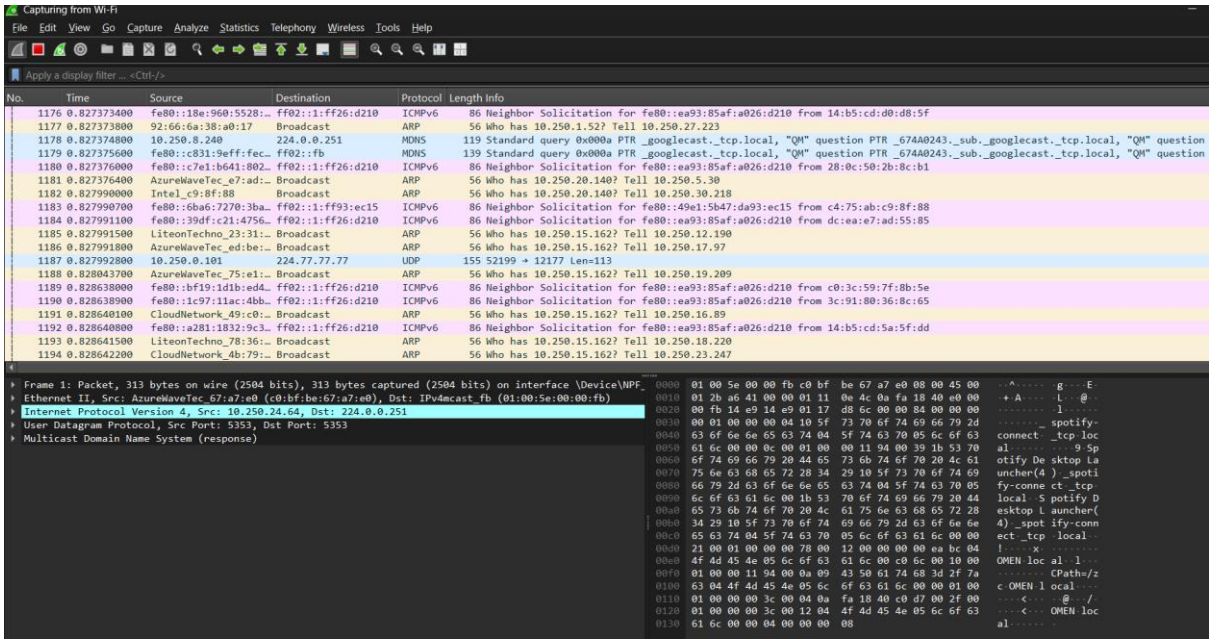
Name : G.Rishitha

Encapsulation Builder: Analyze the overhead introduced by protocol headers and determine how encapsulation affects network efficiency.

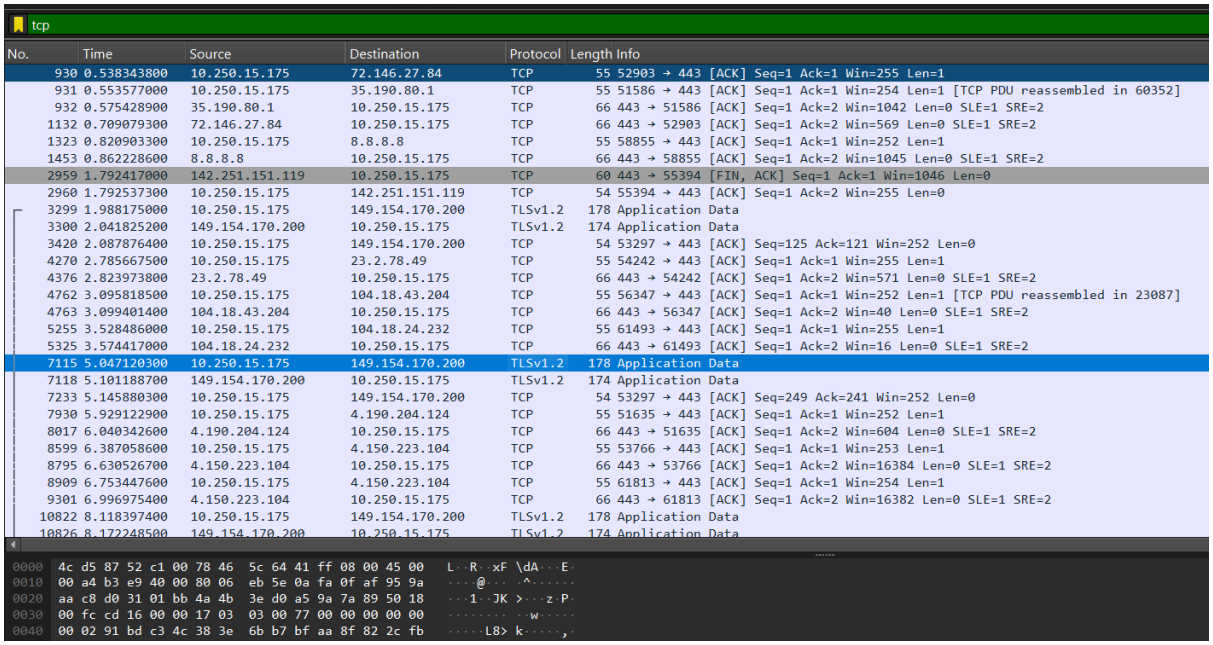
Aim: To capture network packets and understand how Ethernet, IP, and TCP/UDP headers are added during data transmission. Whenever data travels through a computer network, each protocol layer adds its own header to the data. These headers introduce overhead. Using Wireshark, we will capture real network packets, examine Ethernet, IP and TCP/UDP headers, calculate the total overhead, and determine how efficiently the network carries application data



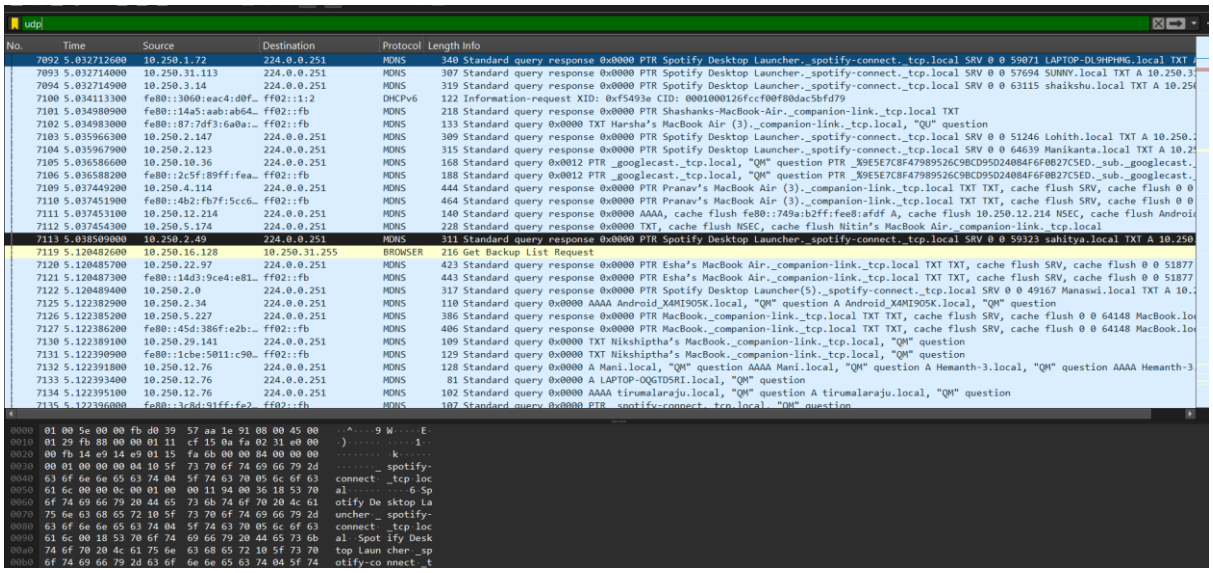

The Wireshark Network Analyzer
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
Apply a display filter ... <Ctrl-/>
Welcome to Wireshark
4.6.9 (v4.6.9-0-g2d548b197c75). You receive automatic updates.
Capture
...using this filter: Enter a capture filter ...
Wi-Fi
Adapter for loopback traffic capture
Local Area Connection* 10
Local Area Connection* 9
Local Area Connection* 8
Bluetooth Network Connection
Local Area Connection* 2
Local Area Connection* 1
vEthernet (WSL (Hyper-V firewall))
Ethernet
Event Tracing for Windows (ETW) reader: etwdump


Capturing from Wi-Fi
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
Apply a display filter ... <Ctrl-/>
No. Time Source Destination Protocol Length Info
1176 0.827373400 fe80::18e:960:5528:... ff02::1:ff26:d210 ICMPv6 86 Neighbor Solicitation for fe80::ea93:85af:a026:d210 from 14:b5:cd:d0:d8:5f
1177 0.827373800 92:66:6a:38:a0:17 Broadcast ARP 56 Who has 10.250.1.52? Tell 10.250.27.223
1178 0.827374800 10.250.8.240 224.0.0.251 MDNS 119 Standard query 0x000a PTR _googlecast._tcp.local, "QM" question PTR _674A0243._sub._googlecast._tcp.local, "QM" question
1179 0.827375600 fe80::c831:9eff:fec... ff02::fb MDNS 139 Standard query 0x000a PTR _googlecast._tcp.local, "QM" question PTR _674A0243._sub._googlecast._tcp.local, "QM" question
1180 0.827376000 fe80::c7e1:b641:802... ff02::1:ff26:d210 ICMPv6 86 Neighbor Solicitation for fe80::ea93:85af:a026:d210 from 28:0c:50:2b:8c:b1
1181 0.827376400 AzureWaveTec_e7:ad:... Broadcast ARP 56 Who has 10.250.20.140? Tell 10.250.5.30
1182 0.827990000 Intel_c9:8f:88 Broadcast ARP 56 Who has 10.250.20.140? Tell 10.250.30.218
1183 0.827990700 fe80::6ba6:7270:3ba... ff02::1:ff93:ec15 ICMPv6 86 Neighbor Solicitation for fe80::49e1:5b47:da93:ec15 from c4:75:ab:c9:8f:88
1184 0.827991100 fe80::39df:c21:4756... ff02::1:ff26:d210 ICMPv6 86 Neighbor Solicitation for fe80::ea93:85af:a026:d210 from dc:ea:e7:ad:55:85
1185 0.827991500 LiteonTechno_23:31:... Broadcast ARP 56 Who has 10.250.15.162? Tell 10.250.12.190
1186 0.827991800 AzureWaveTec_ed:be:... Broadcast ARP 56 Who has 10.250.15.162? Tell 10.250.17.97
1187 0.827992800 10.250.0.101 224.77.77.77 UDP 155 52199 → 12177 Len=113
1188 0.828043700 AzureWaveTec_75:e1:... Broadcast ARP 56 Who has 10.250.15.162? Tell 10.250.19.209
1189 0.828638000 fe80::bf19:1d1b:ed4... ff02::1:ff26:d210 ICMPv6 86 Neighbor Solicitation for fe80::ea93:85af:a026:d210 from c0:3c:59:7f:8b:5e
1190 0.828638900 fe80::1c97:11ac:4bb... ff02::1:ff26:d210 ICMPv6 86 Neighbor Solicitation for fe80::ea93:85af:a026:d210 from 3c:91:80:36:8c:65
1191 0.828640100 CloudNetwork_49:c0:... Broadcast ARP 56 Who has 10.250.15.162? Tell 10.250.16.89
1192 0.828640800 fe80::a281:1832:9c3... ff02::1:ff26:d210 ICMPv6 86 Neighbor Solicitation for fe80::ea93:85af:a026:d210 from 14:b5:cd:5a:5f:dd
1193 0.828641500 LiteonTechno_78:36:... Broadcast ARP 56 Who has 10.250.15.162? Tell 10.250.18.220
1194 0.828642200 CloudNetwork_4b:79:... Broadcast ARP 56 Who has 10.250.15.162? Tell 10.250.23.247
Frame 1: Packet, 313 bytes on wire (2504 bits), 313 bytes captured (2504 bits) on interface \Device\NPF_
Ethernet II, Src: AzureWaveTec_67:a7:e0 (c0:bf:be:67:a7:e0), Dst: IPv4mcast_fb (01:00:5e:00:00:fb)
Internet Protocol Version 4, Src: 10.250.24.64, Dst: 224.0.0.251
User Datagram Protocol, Src Port: 5353, Dst Port: 5353
Multicast Domain Name System (response)

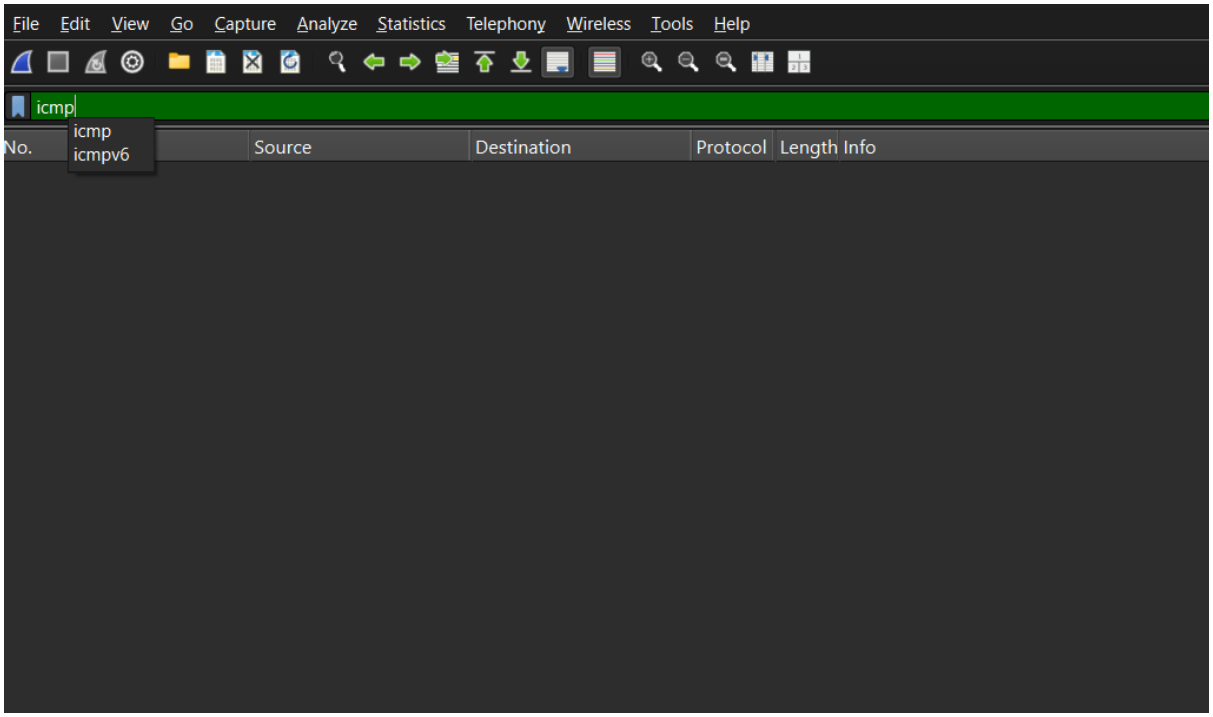
TCP :


tcp
No. Time Source Destination Protocol Length Info
930 0.538343800 10.250.15.175 72.146.27.84 TCP 55 52903 → 443 [ACK] Seq=1 Ack=1 Win=255 Len=1
931 0.553577000 10.250.15.175 35.190.80.1 TCP 55 51586 → 443 [ACK] Seq=1 Ack=1 Win=254 Len=1 [TCP PDU reassembled in 60352]
932 0.575428900 35.190.80.1 10.250.15.175 TCP 66 443 → 51586 [ACK] Seq=1 Ack=2 Win=1042 Len=0 SLE=1 SRE=2
1132 0.709079300 72.146.27.84 10.250.15.175 TCP 66 443 → 52903 [ACK] Seq=1 Ack=2 Win=569 Len=0 SLE=1 SRE=2
1323 0.820903300 10.250.15.175 8.8.8.8 TCP 55 58855 → 443 [ACK] Seq=1 Ack=1 Win=252 Len=1
1453 0.862228600 8.8.8.8 10.250.15.175 TCP 66 443 → 58855 [ACK] Seq=1 Ack=2 Win=1045 Len=0 SLE=1 SRE=2
2959 1.792417000 142.251.151.119 10.250.15.175 TCP 60 443 → 55394 [FIN, ACK] Seq=1 Ack=1 Win=1046 Len=0
2960 1.792537300 10.250.15.175 142.251.151.119 TCP 54 55394 → 443 [ACK] Seq=1 Ack=2 Win=255 Len=0
3299 1.988175000 10.250.15.175 149.154.170.200 TLSv1.2 178 Application Data
3300 2.041825200 149.154.170.200 10.250.15.175 TLSv1.2 174 Application Data
3420 2.087876400 10.250.15.175 149.154.170.200 TCP 54 53297 → 443 [ACK] Seq=125 Ack=121 Win=252 Len=0
4270 2.785667500 10.250.15.175 23.2.78.49 TCP 55 54242 → 443 [ACK] Seq=1 Ack=1 Win=255 Len=1
4376 2.823973800 23.2.78.49 10.250.15.175 TCP 66 443 → 54242 [ACK] Seq=1 Ack=2 Win=571 Len=0 SLE=1 SRE=2
4762 3.095818500 10.250.15.175 104.18.43.204 TCP 55 56347 → 443 [ACK] Seq=1 Ack=1 Win=252 Len=1 [TCP PDU reassembled in 23087]
4763 3.099401400 104.18.43.204 10.250.15.175 TCP 66 443 → 56347 [ACK] Seq=1 Ack=2 Win=40 Len=0 SLE=1 SRE=2
5255 3.528486000 10.250.15.175 104.18.24.232 TCP 55 61493 → 443 [ACK] Seq=1 Ack=1 Win=255 Len=1
5325 3.574417000 104.18.24.232 10.250.15.175 TCP 66 443 → 61493 [ACK] Seq=1 Ack=2 Win=16 Len=0 SLE=1 SRE=2
7115 5.047120300 10.250.15.175 149.154.170.200 TLSv1.2 178 Application Data
7118 5.101188700 149.154.170.200 10.250.15.175 TLSv1.2 174 Application Data
7233 5.145880300 10.250.15.175 149.154.170.200 TCP 54 53297 → 443 [ACK] Seq=249 Ack=241 Win=252 Len=0
7930 5.929122900 10.250.15.175 4.190.204.124 TCP 55 51635 → 443 [ACK] Seq=1 Ack=1 Win=252 Len=1
8017 6.040342600 4.190.204.124 10.250.15.175 TCP 66 443 → 51635 [ACK] Seq=1 Ack=2 Win=604 Len=0 SLE=1 SRE=2
8599 6.387058600 10.250.15.175 4.150.223.104 TCP 55 53766 → 443 [ACK] Seq=1 Ack=1 Win=253 Len=1
8795 6.630526700 4.150.223.104 10.250.15.175 TCP 66 443 → 53766 [ACK] Seq=1 Ack=2 Win=16384 Len=0 SLE=1 SRE=2
8909 6.753447600 10.250.15.175 4.150.223.104 TCP 55 61813 → 443 [ACK] Seq=1 Ack=1 Win=254 Len=1
9301 6.996975400 4.150.223.104 10.250.15.175 TCP 66 443 → 61813 [ACK] Seq=1 Ack=2 Win=16382 Len=0 SLE=1 SRE=2
10822 8.118397400 10.250.15.175 149.154.170.200 TLSv1.2 178 Application Data
10826 8.172248500 149.154.170.200 10.250.15.175 TLSv1.2 174 Application Data
0000 4c d5 87 52 c1 00 78 46 5c 64 41 ff 08 00 45 00 L··R··xF \dA···E·
0010 00 a4 b3 e9 40 00 80 06 eb 5e 0a fa 0f af 95 9a ····@··· ·^·····
0020 aa c8 d0 31 01 bb 4a 4b 3e d0 a5 9a 7a 89 50 18 ···1··JK >···z·P·
0030 00 fc cd 16 00 00 17 03 03 00 77 00 00 00 00 00 ·········· ··w·····
0040 00 02 91 bd c3 4c 38 3e 6b b7 bf aa 8f 82 2c fb ·····L8> k·····,·

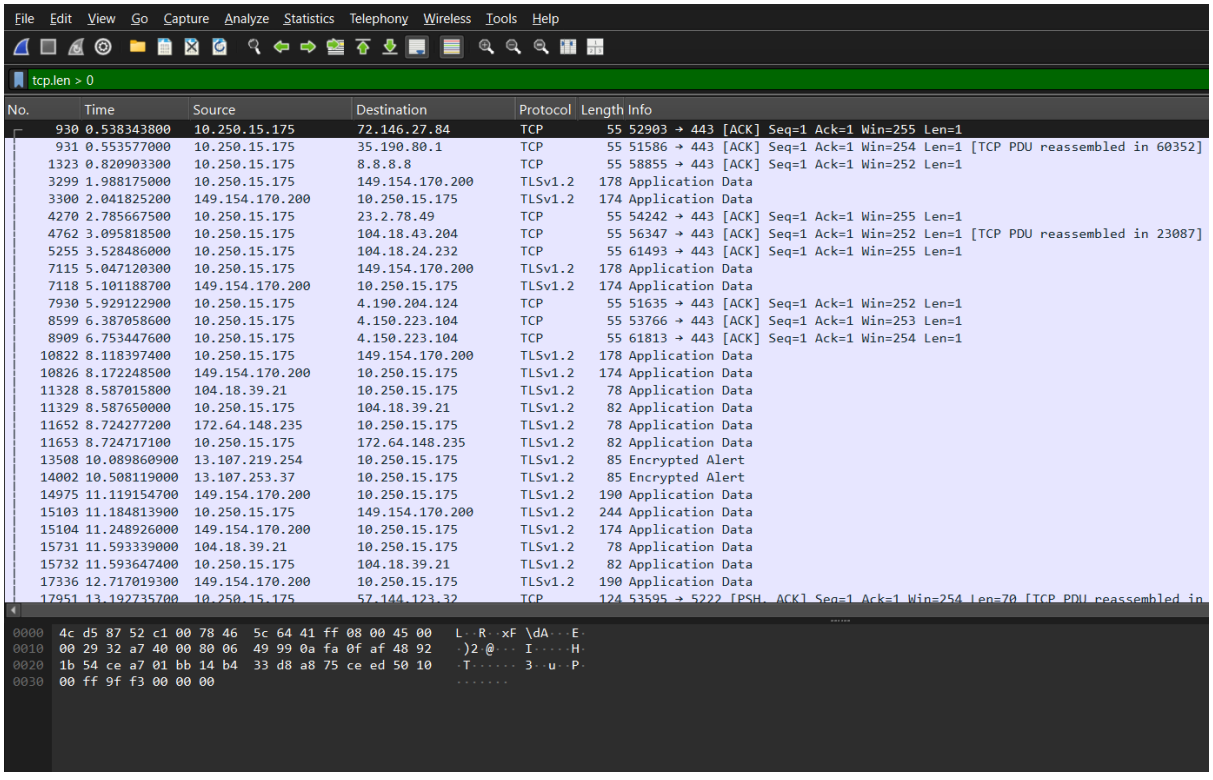
UDP:

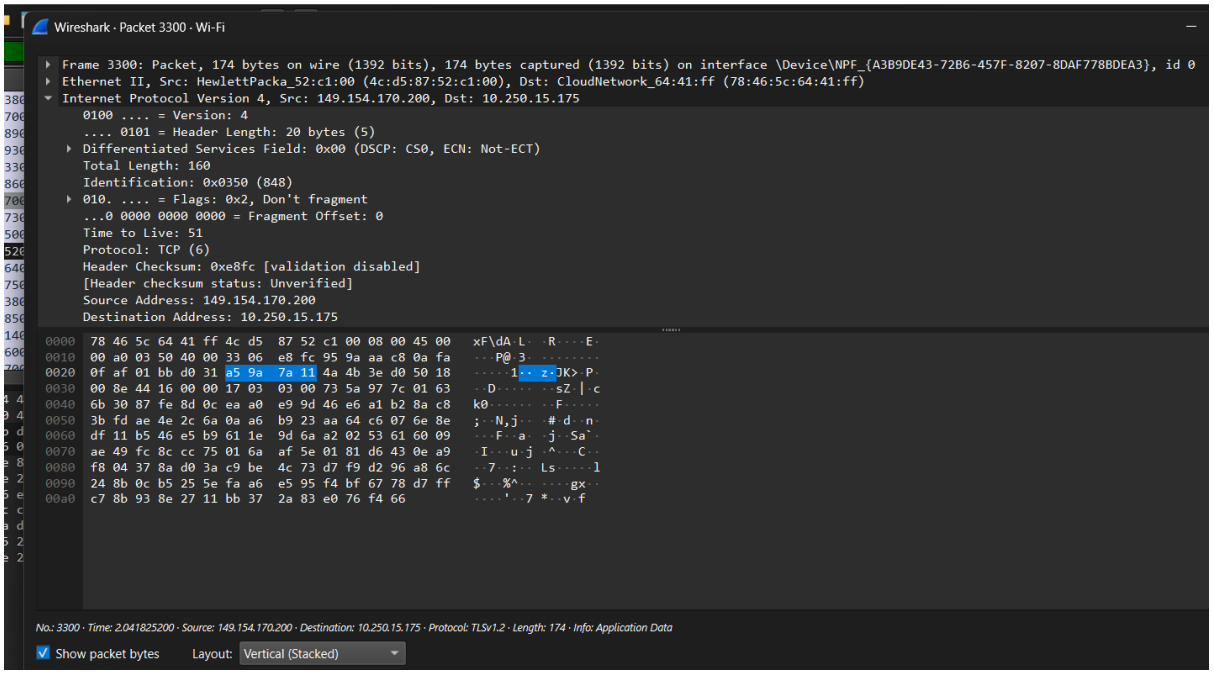

udp
No. Time Source Destination Protocol Length Info
7092 5.032712600 10.250.1.72 224.0.0.251 MDNS 340 Standard query response 0x0000 PTR Spotify Desktop Launcher._spotify-connect._tcp.local SRV 0 0 59071 LAPTOP-DL9HPHMG.local TXT
7093 5.032714000 10.250.31.113 224.0.0.251 MDNS 307 Standard query response 0x0000 PTR Spotify Desktop Launcher._spotify-connect._tcp.local SRV 0 0 57694 SUNNY.local TXT A 10.250.3
7094 5.032714900 10.250.3.14 224.0.0.251 MDNS 319 Standard query response 0x0000 PTR Spotify Desktop Launcher._spotify-connect._tcp.local SRV 0 0 63115 shaikshu.local TXT A 10.25
7100 5.034113300 fe80::3060:eac4:d0f… ff02::1:2 DHCPv6 122 Information-request XID: 0xf5493e CID: 0001000126fccf00f80dac5bfd79
7101 5.034980900 fe80::14a5:aab:ab64… ff02::fb MDNS 218 Standard query response 0x0000 PTR Shashanks-MacBook-Air._companion-link._tcp.local TXT
7102 5.034983000 fe80::87:7df3:6a0a:… ff02::fb MDNS 133 Standard query 0x0000 TXT Harsha's MacBook Air (3)._companion-link._tcp.local, "QU" question
7103 5.035966300 10.250.2.147 224.0.0.251 MDNS 309 Standard query response 0x0000 PTR Spotify Desktop Launcher._spotify-connect._tcp.local SRV 0 0 51246 Lohith.local TXT A 10.250.2
7104 5.035967900 10.250.2.123 224.0.0.251 MDNS 315 Standard query response 0x0000 PTR Spotify Desktop Launcher._spotify-connect._tcp.local SRV 0 0 64639 Manikanta.local TXT A 10.25
7105 5.036586600 10.250.10.36 224.0.0.251 MDNS 168 Standard query 0x0012 PTR _googlecast._tcp.local, "QM" question PTR _%9E5E7C8F47989526C9BCD95D24084F6F0B27C5ED._sub._googlecast.
7106 5.036588200 fe80::2c5f:89ff:fea… ff02::fb MDNS 188 Standard query 0x0012 PTR _googlecast._tcp.local, "QM" question PTR _%9E5E7C8F47989526C9BCD95D24084F6F0B27C5ED._sub._googlecast.
7109 5.037449200 10.250.4.114 224.0.0.251 MDNS 444 Standard query response 0x0000 PTR Pranav's MacBook Air (3)._companion-link._tcp.local TXT TXT, cache flush SRV, cache flush 0 0
7110 5.037451900 fe80::4b2:fb7f:5cc6… ff02::fb MDNS 464 Standard query response 0x0000 PTR Pranav's MacBook Air (3)._companion-link._tcp.local TXT TXT, cache flush SRV, cache flush 0 0
7111 5.037453100 10.250.12.214 224.0.0.251 MDNS 140 Standard query response 0x0000 AAAA, cache flush fe80::749a:b2ff:fee8:afdf A, cache flush 10.250.12.214 NSEC, cache flush Androi
7112 5.037454300 10.250.5.174 224.0.0.251 MDNS 228 Standard query response 0x0000 TXT, cache flush NSEC, cache flush Nitin's MacBook Air._companion-link._tcp.local
7113 5.038509000 10.250.2.49 224.0.0.251 MDNS 311 Standard query response 0x0000 PTR Spotify Desktop Launcher._spotify-connect._tcp.local SRV 0 0 59323 sahitya.local TXT A 10.250
7119 5.120482600 10.250.16.128 10.250.31.255 BROWSER 216 Get Backup List Request
7120 5.120485700 10.250.22.97 224.0.0.251 MDNS 423 Standard query response 0x0000 PTR Esha's MacBook Air._companion-link._tcp.local TXT TXT, cache flush SRV, cache flush 0 0 51877
7121 5.120487300 fe80::14d3:9ce4:e81… ff02::fb MDNS 443 Standard query response 0x0000 PTR Esha's MacBook Air._companion-link._tcp.local TXT TXT, cache flush SRV, cache flush 0 0 51877
7122 5.120489400 10.250.2.0 224.0.0.251 MDNS 317 Standard query response 0x0000 PTR Spotify Desktop Launcher(5)._spotify-connect._tcp.local SRV 0 0 49167 Manaswi.local TXT A 10.2
7125 5.122382900 10.250.2.34 224.0.0.251 MDNS 110 Standard query 0x0000 AAAA Android_X4MI9O5K.local, "QM" question A Android_X4MI9O5K.local, "QM" question
7126 5.122385200 10.250.5.227 224.0.0.251 MDNS 386 Standard query response 0x0000 PTR MacBook._companion-link._tcp.local TXT TXT, cache flush SRV, cache flush 0 0 64148 MacBook.lo
7127 5.122386200 fe80::45d:386f:e2b:… ff02::fb MDNS 406 Standard query response 0x0000 PTR MacBook._companion-link._tcp.local TXT TXT, cache flush SRV, cache flush 0 0 64148 MacBook.lo
7130 5.122389100 10.250.29.141 224.0.0.251 MDNS 109 Standard query 0x0000 TXT Nikshiptha's MacBook._companion-link._tcp.local, "QM" question
7131 5.122390900 fe80::1cbe:5011:c90… ff02::fb MDNS 129 Standard query 0x0000 TXT Nikshiptha's MacBook._companion-link._tcp.local, "QM" question
7132 5.122391800 10.250.12.76 224.0.0.251 MDNS 128 Standard query 0x0000 A Mani.local, "QM" question AAAA Mani.local, "QM" question A Hemanth-3.local, "QM" question AAAA Hemanth-3
7133 5.122393400 10.250.12.76 224.0.0.251 MDNS 81 Standard query 0x0000 A LAPTOP-OQGTD5RI.local, "QM" question
7134 5.122395100 10.250.12.76 224.0.0.251 MDNS 102 Standard query 0x0000 AAAA tirumalaraju.local, "QM" question A tirumalaraju.local, "QM" question
7135 5.122396000 fe80::3c8d:91ff:fe2… ff02::fb MDNS 107 Standard query 0x0000 PTR _spotify-connect._tcp.local, "QM" question
0000 01 00 5e 00 00 fb d0 39 57 aa 1e 91 08 00 45 00 ··^····9 W·····E·
0010 01 29 fb 88 00 00 01 11 cf 15 0a fa 02 31 e0 00 ·)·····1··
0020 00 fb 14 e9 14 e9 01 15 fa 6b 00 00 84 00 00 00 ·····k····
0030 00 01 00 00 00 04 10 5f 73 70 6f 74 69 66 79 2d ·····_ spotify-
0040 63 6f 6e 6e 65 63 74 04 5f 74 63 70 05 6c 6f 63 connect· _tcp·loc
0050 61 6c 00 00 0c 00 01 00 00 11 94 00 36 18 53 70 al····· ····6·Sp
0060 6f 74 69 66 79 20 44 65 73 6b 74 6f 70 20 4c 61 otify De sktop La
0070 75 6e 63 68 65 72 10 5f 73 70 6f 74 69 66 79 2d uncher· _spotify-
0080 63 6f 6e 6e 65 63 74 04 5f 74 63 70 05 6c 6f 63 connect· _tcp·loc
0090 61 6c 00 18 53 70 6f 74 69 66 79 20 44 65 73 6b al··Spot ify Desk
00a0 74 6f 70 20 4c 61 75 6e 63 68 65 72 10 5f 73 70 top Laun cher·_sp
00b0 6f 74 69 66 79 2d 63 6f 6e 6e 65 63 74 04 5f 74 otify-co nnect·_t

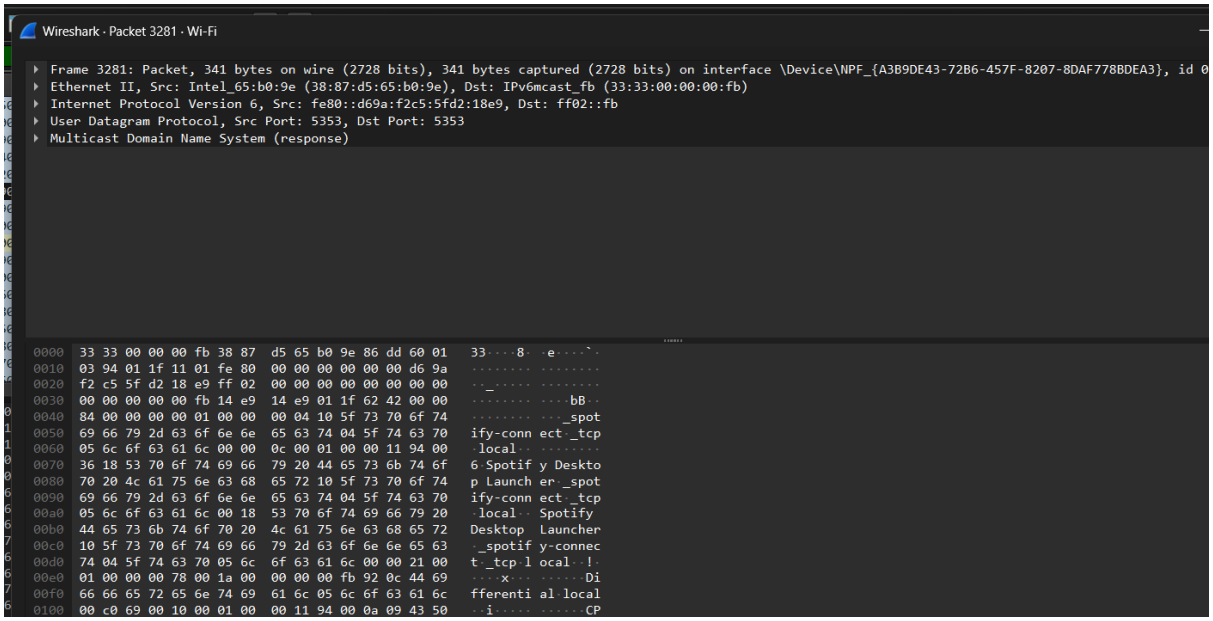
ICMP:


File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
icmp
icmp
icmpv6
No.
Source
Destination
Protocol
Length Info

tcp.len > 0:


File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
tcp.len > 0
No. Time Source Destination Protocol Length Info
930 0.538343800 10.250.15.175 72.146.27.84 TCP 55 52903 → 443 [ACK] Seq=1 Ack=1 Win=255 Len=1
931 0.553577000 10.250.15.175 35.190.80.1 TCP 55 51586 → 443 [ACK] Seq=1 Ack=1 Win=254 Len=1 [TCP PDU reassembled in 60352]
1323 0.820903300 10.250.15.175 8.8.8.8 TCP 55 58855 → 443 [ACK] Seq=1 Ack=1 Win=252 Len=1
3299 1.988175000 10.250.15.175 149.154.170.200 TLSv1.2 178 Application Data
3300 2.041825200 149.154.170.200 10.250.15.175 TLSv1.2 174 Application Data
4270 2.785667500 10.250.15.175 23.2.78.49 TCP 55 54242 → 443 [ACK] Seq=1 Ack=1 Win=255 Len=1
4762 3.095818500 10.250.15.175 104.18.43.204 TCP 55 56347 → 443 [ACK] Seq=1 Ack=1 Win=252 Len=1 [TCP PDU reassembled in 23087]
5255 3.528486000 10.250.15.175 104.18.24.232 TCP 55 61493 → 443 [ACK] Seq=1 Ack=1 Win=255 Len=1
7115 5.047120300 10.250.15.175 149.154.170.200 TLSv1.2 178 Application Data
7118 5.101188700 149.154.170.200 10.250.15.175 TLSv1.2 174 Application Data
7930 5.929122900 10.250.15.175 4.190.204.124 TCP 55 51635 → 443 [ACK] Seq=1 Ack=1 Win=252 Len=1
8599 6.387058600 10.250.15.175 4.150.223.104 TCP 55 53766 → 443 [ACK] Seq=1 Ack=1 Win=253 Len=1
8909 6.753447600 10.250.15.175 4.150.223.104 TCP 55 61813 → 443 [ACK] Seq=1 Ack=1 Win=254 Len=1
10822 8.118397400 10.250.15.175 149.154.170.200 TLSv1.2 178 Application Data
10826 8.172248500 149.154.170.200 10.250.15.175 TLSv1.2 174 Application Data
11328 8.587015800 104.18.39.21 10.250.15.175 TLSv1.2 78 Application Data
11329 8.587650000 10.250.15.175 104.18.39.21 TLSv1.2 82 Application Data
11652 8.724277200 172.64.148.235 10.250.15.175 TLSv1.2 78 Application Data
11653 8.724717100 10.250.15.175 172.64.148.235 TLSv1.2 82 Application Data
13508 10.089860900 13.107.219.254 10.250.15.175 TLSv1.2 85 Encrypted Alert
14002 10.508119000 13.107.253.37 10.250.15.175 TLSv1.2 85 Encrypted Alert
14975 11.119154700 149.154.170.200 10.250.15.175 TLSv1.2 190 Application Data
15103 11.184813900 10.250.15.175 149.154.170.200 TLSv1.2 244 Application Data
15104 11.248926000 149.154.170.200 10.250.15.175 TLSv1.2 174 Application Data
15731 11.593339000 104.18.39.21 10.250.15.175 TLSv1.2 78 Application Data
15732 11.593647400 10.250.15.175 104.18.39.21 TLSv1.2 82 Application Data
17336 12.717019300 149.154.170.200 10.250.15.175 TLSv1.2 190 Application Data
17951 13.192735700 10.250.15.175 57.144.123.32 TCP 124 53595 → 5222 [PSH, ACK] Seq=1 Ack=1 Win=254 Len=70 [TCP PDU reassembled in
0000 4c d5 87 52 c1 00 78 46 5c 64 41 ff 08 00 45 00
0010 00 29 32 a7 40 00 80 06 49 99 0a fa 0f af 48 92
0020 1b 54 ce a7 01 bb 14 b4 33 d8 a8 75 ce ed 50 10
0030 00 ff 9f f3 00 00 00





Step 6: Calculate Header Overhead

Calculate the header overhead using the actual packets captured in Wireshark.

1. TCP Packet

Given:

- Ethernet Header = 14 bytes
- IPv4 Header = 20 bytes
- TCP Header = 20 bytes
- Application Data = 0 bytes

Total Size:

14 + 20 + 20 + 0 = 54 bytes

Total Overhead:

14 + 20 + 20 = 54 bytes

Payload Efficiency:

(0 / 54) × 100 = 0%

2. UDP Packet

Given:

- Ethernet Header = 14 bytes
- IPv4 Header = 20 bytes
- UDP Header = 8 bytes
- Application Data = 303 bytes

Total Size:

14 + 20 + 8 + 303 = 345 bytes

Total Overhead:

14 + 20 + 8 = 42 bytes

Payload Efficiency:

(303 / 345) × 100 = 87.83%

Note: Wireshark may not capture or display the Ethernet FCS. The calculations are based on the actual captured packet lengths.

Step 7:

Parameter	**TCP Packet**	**UDP Packet**
Frame Length	54 bytes	345 bytes
Ethernet Header Length	14 bytes	14 bytes
IP Header Length	20 bytes	20 bytes
Transport Header Length	20 bytes	8 bytes
IP Total Length	40 bytes	331 bytes
Payload Length	0 bytes	303 bytes